

Low-Cost Differential Impedance AST Prototype: Complete Buildable Design and Validation Protocol

TL;DR

- **A practical, benchtop differential-impedance AST prototype can be built for roughly \$350–\$700 (turnkey eval boards) or \$150–\$300 (bare-IC custom build) using an Analog Devices AD5941-based impedance analog front end (bare AD5941BCPZ IC ~\$14, EVAL-AD5941 daughterboard ~\$205, or EVAL-AD5940BIOZ shield ~\$439), or an AD5933 board (EVAL-AD5933EBZ ~\$68 at DigiKey), plus commercial screen-printed interdigitated gold electrodes (DropSens/Metrohm G-IDEAU10, quote-**

only) or DIY ENIG-gold PCB electrodes, a microcontroller, a ~\$30–\$100 thermostatic heater, and disposable low-volume wells – one to three orders of magnitude cheaper and far simpler than the SiNWFET lab-on-silicon-chip (LOSC) in the uploaded paper, which requires cleanroom EBL/RIE, ALD HfO₂, and an HP4155A semiconductor parameter analyzer.

- **The paper's two acceleration strategies translate to impedance sensing with one critical inversion:** low-ionic-strength media accelerates detection even *more* for impedance than for pH because impedance sensitivity scales inversely with baseline medium conductivity – but the paper's 0.9% NaCl (high conductivity) is the *wrong* choice for impedance; use a HEPES-based low-conductivity zwitterionic growth buffer or diluted broth instead. Published work (Swami et al. 2022; Karmakar/Gopalakrishnan et al. 2025) shows low-conductivity media enables 20–80 min impedance AST. Volumetric cell

concentration also helps but adds cost;
skip picoliter microfluidics for v1.

- **Set the susceptibility call using a biological-replicate statistical threshold directly analogous to the paper's $\Delta V_{th} > 20$ mV (= max SEM) rule:** define the noise floor as mean + 3×SD of the sterility control's impedance fluctuation, compute the Differential Impedance Ratio $DIR(t) = \Delta Z_{antibiotic} / \Delta Z_{growth-control}$ (≈ 1 = resistant, ≈ 0 = susceptible), and extract an electrical-MIC where normalized conductivity change falls below threshold – validated against E-test/broth microdilution MICs.

Key Findings

1. **Hardware:** The AD5933 (1 kHz–100 kHz sweep, 12-bit, DFT-based, EVAL-AD5933EBZ ~\$68 at DigiKey) and the newer AD5941/AD5940 (~0.015 Hz–200 kHz, 16-bit, 4-wire, integrated potentiostat/TIA) are the two dominant low-cost bioimpedance front ends. The

AD5941 is the better fit for bacterial growth impedance because it reaches the low frequencies where medium-conductivity change dominates, offers 4-wire measurement to suppress electrode polarization drift over hours-long runs, and has higher resolution. The AD5933's native 1 kHz floor can be extended down to ~10 Hz with external analog front-end circuitry, as multiple published groups have done.

2. **Electrodes:** Commercial screen-printed interdigitated gold electrodes (Metrohm DropSens G-IDEAU10, 10 μm lines/gaps; or G-IDEAU5, 5 μm , 250 finger pairs on glass) are the fastest path; DIY PCB ENIG-gold interdigitated electrodes (~150–200 μm features) and gold-leaf/laser-ablated electrodes are viable ultra-low-cost alternatives. Sensitivity is dominated by finger gap: smaller gaps concentrate the field near the surface and raise sensitivity, but a substantial fraction of the field must sample the bulk medium to detect metabolic conductivity changes, so ~5–10

μm gaps are a good compromise for bulk-conductivity sensing.

3. **Acceleration strategy translation:**

Impedance magnitude $\propto 1/(\text{medium conductivity})$, so a low-conductivity medium raises both baseline impedance and the *fractional* change per unit of metabolic ion production – a direct win. The tradeoff versus pH sensing: 0.9% NaCl (~150 mM) is ideal for unbuffered pH dynamic range but is high-conductivity for impedance and swamps the signal; a zwitterionic HEPES buffer or diluted broth (1–10%) gives much higher impedance dynamic range while still supporting 1–2 doubling cycles.

4. **Threshold methodology:** Classic

impedance microbiology (Bactometer, RABIT, Malthus, BacTrac) defines "detection time" as the point where the impedance/conductance curve deviates from baseline by a set threshold (BacTrac uses 5%). Modern impedance AST papers use normalized metrics: Spencer et al.

2023 (ACS Sensors) define the electrical-MIC via a normalized-conductivity threshold; Spencer et al. 2020 (Nat Commun) use a 50% cell-count gate with a t-test.

5. Cost/complexity vs LOSC: The LOSC paper requires SOI wafers, e-beam lithography, reactive-ion etching, ALD HfO₂ gate dielectric, on-chip AgCl/Ag pseudo-reference electrodes, PDMS bonding, and an HP4155A semiconductor parameter analyzer. The impedance prototype needs none of this — an off-the-shelf AFE IC, commercial or PCB electrodes, and an Arduino/Raspberry Pi.

Details

1. Low-Cost Impedance Measurement Hardware

AD5933 (Analog Devices). A single-chip impedance converter combining a DDS sine generator, 12-bit 1-MSPS ADC, and on-chip DSP that returns real and imaginary

impedance via DFT. Native frequency range 1 kHz–100 kHz; measures 1 k Ω –10 M Ω natively. The EVAL-AD5933EBZ is listed at **\$68.24 (qty 1), In-Stock 78, 10-week standard lead time (DigiKey PN 505-EVAL-AD5933EBZ-ND, accessed Aug 2026)**; other distributors quote up to ~\$88. Strengths: cheapest turnkey option; the single most-used bioimpedance IC since 2005; DFT rather than I/Q demodulation. Limitations: the 1 kHz floor sits above the 100 Hz–1 kHz region where medium-conductivity effects are strongest; 2-electrode by default (4-wire needs extra circuitry); 12-bit resolution. Academic groups have added front-end circuits to extend it to ~10 Hz and enable 4-electrode measurement.

AD5941 / AD5940 (Analog Devices) – recommended core. 16-bit 800-kSPS SAR ADC, dual excitation loops, integrated ultra-low-noise potentiostat and transimpedance amplifier, [Mouser](#) native 4-wire impedance, frequency range spanning roughly 0.015 Hz to 200 kHz. **DigiKey pricing (Aug 2026): bare AD5941BCPZ IC \$13.65 (AD5941BCPZ-RL7 \$14.06); EVAL-AD5941 daughterboard ~\$205;**

EVAL-AD5940BIOZ bio-impedance shield (Arduino form factor) \$439.08; ADICUP3029 Cortex-M3 host ~\$64. The AD5940/41 is recommended because it reaches the low frequencies where metabolic conductivity change is largest, its 4-wire capability suppresses electrode-polarization drift (critical for multi-hour growth monitoring), and Analog Devices publishes open BIA/EDA firmware examples on GitHub. [GitHub](#)

Other options (DigiKey, Aug 2026):
MAX30001EVSYS# bioimpedance/ECG AFE eval kit \$162.21 (optimized for respiration/body impedance, narrow band – less suitable); EVAL-CN0565-ARDZ EIT/impedance Arduino shield \$196.22; EVAL-ADUCM355QSPZ \$223.17 (ARM Cortex-M3 + dual potentiostats + impedance in one chip); AFE4300 (1 kHz–255 kHz, body-composition).

Recommended architecture: AD5941 daughterboard + ADICUP3029 [Mouser Electronics](#) (or Raspberry Pi 4 / ESP32 via SPI) as controller/data logger, with an ADG-series analog multiplexer to sweep multiple wells

(test, growth control, sterility control)
sequentially through one AFE — the same
multiplexing approach Scherer et al. (Lab on a
Chip 2021) used to read many nanoliter
chambers in parallel.

2. Electrode Options

Commercial screen-printed IDEs (fastest path). Metrohm DropSens interdigitated gold electrodes are the reference choice in this literature:

- **G-IDEAU10** (DRP-G-IDEAU10-U20): gold, 10 μm lines/gaps, glass substrate 22.9×7.6 mm, [Metrohm](#) sold in packs of 20.
- **G-IDEAU5**: gold, 5 μm lines/gaps; the G-IDEAU5 has 250 digit pairs, 5 μm width, 6.76 mm length with a well-characterized cell constant in the literature.
- Platinum (G-IDEPT5/10) and cheaper carbon variants also exist. [Metrohm](#) Pricing is quote-only — no public list price; request a quote from Metrohm DropSens (info.dropsens@metrohm.com) or distributor nLab.

Hannah et al. (2019/2020, Strathclyde) built a validated AST on commercial screen-printed *gold* electrodes with an antibiotic-seeded agarose hydrogel, distinguishing susceptible *S. aureus* from MRSA and susceptible/resistant *E. coli* in under 45–60 min – a strong precedent for this exact electrode class. That work has since spun into a commercial venture (Microplate Dx).

DIY PCB gold (ENIG) interdigitated electrodes (lowest cost at volume). Standard PCB fabrication with electroless-nickel-immersion-gold (ENIG) finish yields gold-surfaced interdigitated combs at feature sizes down to ~150–200 μm for a few dollars per board from any PCB house. Published groups have used ENIG PCB electrodes for impedance biosensing (SARS-CoV-2 spike protein, cancer-cell ECIS, Salmonella detection with 200 μm width/spacing). Tradeoff: the nickel underlayer and coarser features reduce sensitivity and add electrochemical background versus pure-gold thin-film IDEs; some groups add supplemental gold electroplating.

Gold-leaf / laser-ablated electrodes (ultra-cheap DIY). 24-karat gold leaf laminated to PVC adhesive and patterned by laser ablation produces reproducible interdigitated electrodes suitable for EIS at very low cost without cleanroom access (demonstrated for E. coli immunodetection).

Geometry guidance. For sensing bulk-medium conductivity changes from metabolism (rather than surface capture), the electric field must penetrate the bulk, and field penetration depth scales with finger pitch. Very small gaps (sub- μm to few- μm) confine the field near the surface and maximize surface-event sensitivity but sample less bulk; larger gaps (10–200 μm) sample more bulk. A 5–10 μm gap gold IDE is the compromise. Simulation and patent literature agree that reducing gap size raises sensitivity, with gap having greater influence on E-field intensity than finger width.

3. Translating the LOSC Paper's Acceleration Strategies to Impedance

Strategy (a) – unbuffered, low-ionic-strength media. In the paper, replacing LB with unbuffered 0.9% NaCl + 1% glucose accelerated the detectable metabolic pH signal from 50–70 min to 20 min by removing buffering capacity so acidic overflow metabolites (acetate, pyruvate, succinate) shift pH fast. **For impedance the physics is even more favorable, but the optimal medium differs.** Because $|Z| \propto 1/(\sigma \cdot \omega \cdot \text{geometry})$:

- A *low-conductivity* medium raises baseline $|Z|$ and, crucially, raises the *fractional* impedance change produced by each increment of metabolically generated ions. This is exactly why Swami et al. (Biosens. Bioelectron. 2022, 200:113876, IIT Delhi, PMID 34974262) introduced a low-conductivity zwitterionic growth buffer: "*a low conductivity zwitterionic buffer is used for boosting impedance sensitivity in simple interdigitated electrodes (IDEs) allowing rapid AST in just 20 min without any liquid flow*"; their LCGB "*allows rapid MIC determination in 80 min and antibiotic sensitivity analysis in just 60 min, which is at*

least 8–9 times faster" than LB, at 2×10^5 CFU. Likewise, Karmakar/Gopalakrishnan et al. (Sci. Rep. 2025, s41598-024-84286-3) chose "a diluted nutrient medium as an electrolyte since it provides a higher charge transfer baseline signal for better sensitivity."

- **The key inversion versus pH sensing:** 0.9% NaCl (~150 mM) is excellent for *unbuffered pH* dynamic range but is *high-conductivity* from an impedance standpoint – the added ions swamp the metabolic signal. For impedance you want to *minimize* background ionic strength: use a HEPES-based zwitterionic low-conductivity buffer (zwitterions carry little net charge, so conductivity is low while osmolarity/pH support growth) or dilute the broth (1–10% LB / diluted Mueller-Hinton). [ScienceDirect](#)
This is the single most important design translation: keep the glucose (metabolic fuel) but replace most of the NaCl with a zwitterionic buffer or dilution to maximize impedance dynamic range.

Strategy (b) – volumetric confinement / cell concentration. The paper concentrated cells ~2 orders of magnitude (to $\sim 3 \times 10^8$ cells/mL) in 35-pL chambers, dramatically boosting local metabolic signal. For impedance this also helps (higher cell density $\rightarrow$ faster measurable conductivity change), and the classic impedance-microbiology "detection time" corresponds to reaching $\sim 10^6$ – 10^7 cells/mL. But microfluidic pre-concentration adds cost and complexity that conflicts with the low-cost goal. **Recommended compromise:** use a small static well (10–100 μ L) with the CLSI-standard starting inoculum of 5×10^5 CFU/mL, accept a somewhat longer time-to-detection (30 min–3 h vs 20 min), and recover speed instead through low-conductivity medium and low-frequency measurement. If faster results are needed, add a simple membrane-filter or centrifugal pre-concentration step rather than picoliter chambers.

Net tradeoff summary: Low-conductivity medium is a free lunch for impedance (cheaper AND faster AND more sensitive).

Volumetric confinement is beneficial but expensive; skip it for v1.

4. Experimental Design / Protocol

Strains (mirroring the paper, using accessible ATCC equivalents). The paper used E. coli MG1655 (susceptible) with isogenic AMP-R (lacIZYA::bla), CIP-R (gyrA), and NIT-R (Δ nfsAB) strains plus UPEC536. For an accessible reference-strain validation:

- **Susceptible reference:** E. coli ATCC 25922 (CLSI/EUCAST QC strain; published ampicillin MIC QC range and breakpoints).
- **Resistant reference:** E. coli ATCC 35218 (TEM-1 β -lactamase producer, ampicillin-resistant; a CLSI/EUCAST QC strain used to check β -lactam/inhibitor components). This provides a clean AMP susceptible/resistant pair from authenticated QC strains; sold ready-to-use (ATCC MINI-PACK, Microbiologics pellets).
- **Gram-positive control:** S. aureus ATCC 29213 (MIC QC strain). [uspto](#)

- Optionally add clinical CIP-R and NIT-R isolates to mirror the paper's full panel.

Antibiotic panel (adapted from paper's AMP/CIP/NIT).

- Ampicillin (β -lactam, bactericidal): the paper used 30 $\mu\text{g/mL}$; test around the E. coli breakpoint (EUCAST $S \leq 8 \text{ mg/L}$) with a dose series (0, 2, 4, 8, 16, 32, 64 mg/L) for eMIC extraction.
- Ciprofloxacin (fluoroquinolone, bactericidal): paper used 2 $\mu\text{g/mL}$ (near the clinical breakpoint); dose series 0.06–4 mg/L .
- Nitrofurantoin: paper used 200 $\mu\text{g/mL}$; dose series around the 64 mg/L breakpoint.

Media. Primary: low-conductivity growth buffer = 1% glucose + minimal NaCl + HEPES-based zwitterionic buffer, OR diluted Mueller-Hinton broth (5–10%). Run a parallel arm in standard cation-adjusted Mueller-Hinton broth (CAMHB) as the CLSI-reference comparator so susceptibility calls remain traceable to standard methods. Justify the low-conductivity

choice by the $1/\sigma$ impedance-sensitivity argument above.

Inoculum. 0.5 McFarland adjusted, diluted to the CLSI-standard final 5×10^5 CFU/mL (the density used in reference broth microdilution and in Spencer's single-cell work).

Well/replicate structure per antibiotic per strain:

- *Test wells:* inoculum + antibiotic at each concentration.
- *Growth control:* inoculum, no antibiotic (defines maximal ΔZ , the denominator of DIR).
- *Sterility control:* medium + antibiotic, no inoculum (defines noise floor; corrects for abiotic drift/evaporation).
- *Biological triplicates* minimum (the paper used triplicates and defined its threshold from replicate SEM); ideally $n \geq 3$ on separate days.

Incubation / evaporation control. 35–37 °C (CLSI standard). Because conductivity changes $\sim 2\%/^{\circ}\text{C}$, temperature must be actively

controlled and logged (use the AD5933/AD5941 on-chip temperature sensor or a dedicated NTC/thermistor per well); a PID-controlled resistive heater or heating block suffices. Prevent evaporation with a mineral-oil overlay (as the paper used oil sealing) or a sealed humidified chamber — the sterility control differentially subtracts residual drift.

Frequency selection protocol. At assay start, sweep 100 Hz–1 MHz on a growth control vs sterility control and identify the frequency with the maximum $|\Delta Z|$ separation over the first 1–2 h (typically 100 Hz–10 kHz, where bulk-conductivity effects dominate; classic commercial systems operate 2–10 kHz with capacitance changes most pronounced near 2 kHz). Lock that single optimal frequency for high-cadence monitoring.

Measurement cadence and run length. Log $|Z|$ and phase (or R and C) at the locked frequency every 1–5 min for up to 3–4 h. Death-based (bactericidal) signatures can appear in 20–90 min; growth-based signatures in 1–3 h in low-conductivity medium.

5. Quantitative Impedance Threshold for "Evidence of Antibiotic Response"

Baseline noise floor (analogous to the paper's max-SEM rule). The uploaded paper set $\Delta V_{th} > 20$ mV – the maximum SEM across biological replicates – as its fixed susceptibility threshold. The impedance analog: compute the sterility control's impedance fluctuation over the run and define the **noise floor = mean + 3×SD** of that fluctuation (equivalently, use the maximum inter-replicate SEM of the growth-control baseline). Any ΔZ excursion must exceed this to count as real.

Differential metrics.

- $\Delta Z(t) = Z_{\text{test}}(t) - Z_{\text{sterility-control}}(t)$ (abiotic-corrected signal).
- Normalized impedance change:
 $\Delta Z_{\text{norm}}(t) = [Z(t) - Z(0)] / [Z(0) - Z_{\text{empty}}]$, following the normalized-resistance convention of Yang/Ekinci (2020, PNAS) and the normalized-conductivity $\Delta \kappa$ of Spencer 2023.

- **Differential Impedance Ratio: DIR(t) = $\Delta Z_{\text{antibiotic}}(t) / \Delta Z_{\text{growth-control}}(t)$.**

DIR $\approx$ 1 $\rightarrow$ the drug did not inhibit metabolism $\rightarrow$ **resistant**; DIR $\approx$ 0 $\rightarrow$ metabolism suppressed $\rightarrow$ **susceptible**.

This mirrors the resistant/susceptible ratio $\Delta I_{\text{ABX}} / \Delta I_{\text{control}} > T_R$ used in optical AST (where a resistance threshold near 0.5 is applied).

Time-to-Detection (TTD). The time for the growth-control ΔZ to first cross the noise floor (mean + 3 $\times$ SD). Report susceptibility only once the growth control has crossed TTD (proving the assay is live) AND the DIR has stabilized above/below the call threshold.

Susceptibility call threshold. Set a DIR cutoff from the separation of susceptible vs resistant reference pairs across biological replicates — choose the DIR value between the resistant cluster (near 1) and susceptible cluster (near 0) that maximizes categorical agreement, requiring it to exceed the replicate-SEM band (the direct analog of the paper's "20 mV = max SEM" logic). Published precedents: BacTrac

calls detection at a 5% impedance-curve deviation; Safavieh et al. reported (per Spencer et al. ACS Sensors 2023) that *"the medium resistance of an E. coli suspension or a methicillin-resistant Staphylococcus aureus (MRSA) suspension reduced by >50% within 1 h, when the initial bacterial concentration was 10⁶ CFU/mL,"* enabling an end-point susceptibility call after 90 min.

Electrical-MIC extraction and validation

(analogous to the paper's Fig. S18 dose-

response). For each antibiotic, plot the fixed-time (e.g., 60 min) normalized

conductivity/impedance change versus

log(antibiotic concentration). Spencer, Li, Zhu,

Sutton & Morgan ("Electrical Broth Micro-

Dilution for Rapid Antibiotic Resistance

Testing," ACS Sensors 2023, 8(3):1101-1108,

Univ. Southampton/UKHSA/KCL) define the

electrical-MIC (eMIC) from this curve:

normalized conductivity *"falls below 50% at the*

MIC," and *"the data demonstrate the utility of a*

fast electrical-MIC test, where setting the

threshold for conductivity to, e.g., 10% would

correctly identify the MIC within +/- twofold

dilution" (normalized at 60 min, n = 3 biological repeats, 100 nL measurement volume). **Adopt this convention: define eMIC as the lowest concentration where normalized conductivity change drops below the chosen threshold (~50% for the nominal MIC, with a stricter ~10% cutoff giving $\pm$ one two-fold-dilution accuracy).** Validate by linear regression of eMIC against E-test/reference broth-microdilution MIC across the panel and report R^2 — the uploaded paper reported $R^2 = 0.99$ for its pH-based MIC-equivalent vs E-test; target comparable linearity ($R^2 > 0.9$). Also report a "fast-criterion" eMIC at a fixed early timepoint, as the paper did with its 5-minute criterion.

6. Cost Estimate (Bill of Materials)

Low-cost differential impedance AST prototype (benchtop, single-AFE, multiplexed multi-well):

Component	Choice	Approx. cost (USD, Aug 2026)
Impedance AFE	EVAL-AD5941 daughterboard (\$205), or bare AD5941BCPZ IC (\$13.65) for a custom PCB	\$14–\$205
Host controller	ADICUP3029 (\$64), Raspberry Pi 4 (\$55), or ESP32 (~\$10)	\$10–\$64
Electrodes	DropSens G-IDEAU10 gold IDE (quote-only), or DIY ENIG PCB IDEs	~\$2–\$25 each
Analog multiplexer	ADG-series for multi-well sequencing	\$5–\$20

Component	Choice	Approx. cost (USD, Aug 2026)
Temperature control	PID heater block + NTC/thermistors	\$30– \$100
Evaporation control	Mineral-oil overlay + sealed humidified chamber	\$10– \$30
Disposable wells / simple microfluidic	PDMS or laser-cut acrylic wells, 10–100 μ L	\$5–\$50
Misc (wiring, enclosure, PSU)	—	\$30– \$80
Total		$\approx$ \$350– \$700 (turnkey eval)

Component	Choice	Approx. cost (USD, Aug 2026)
		boards) or \$150– \$300 (bare-IC custom build)

Consumables per test (electrode + media + antibiotic) are on the order of a few dollars with DIY PCB electrodes, higher with commercial gold IDEs.

Comparison to the SiNWFET LOSC in the paper. The LOSC approach requires: SOI wafers; electron-beam lithography and reactive-ion etching (cleanroom, high \$/hour); ALD HfO₂ gate dielectric; on-chip AgCl/Ag pseudo-reference electrodes; PDMS microfluidics with O₂-plasma bonding; picoliter chamber arrays; and readout on an HP4155A

semiconductor parameter analyzer (well into five figures used/refurbished, six figures new). Total capital and per-chip fabrication cost is one to three orders of magnitude above the impedance prototype, and it demands cleanroom access and specialist fabrication skills. The impedance prototype trades the LOSC's 20-minute, single-chamber, single-cell-sensitivity performance for a buildable, field-deployable system at hobbyist/small-lab cost – at the price of somewhat longer time-to-result (30 min–3 h) and lower single-cell resolution.

Recommendations

Stage 1 – Bench bring-up (weeks 1–3). Buy one EVAL-AD5941 + ADICUP3029 and a pack of DropSens G-IDEAU10 gold IDEs. Validate the measurement chain against known resistors/RC dummy cells and against KCl solutions of known conductivity. Confirm you can resolve $<1\%$ $|Z|$ changes at 100 Hz–10 kHz. *Go/no-go:* stable baseline drift below the

intended noise floor (target SD < 0.5% over 3 h at fixed T).

Stage 2 – Medium and frequency optimization (weeks 3–6). With *E. coli* ATCC 25922 (growth control vs sterility control), compare CAMHB, diluted MHB (5–10%), and a HEPES low-conductivity growth buffer. Run 100 Hz–1 MHz sweeps and pick the frequency and medium maximizing early $|\Delta Z|$ separation. *Threshold to advance:* growth control crosses the mean+3×SD noise floor within <2 h in the chosen low-conductivity medium.

Stage 3 – Susceptible/resistant discrimination (weeks 6–10). Run the ATCC 25922 (S) vs ATCC 35218 (AMP-R) pair with ampicillin at breakpoint concentration in biological triplicate. Compute DIR(t). *Success:* clear DIR separation (resistant near 1, susceptible near 0) exceeding the replicate-SEM band, with categorical agreement vs disk/E-test on all replicates.

Stage 4 – Dose-response and eMIC validation (weeks 10–16). Generate normalized-impedance dose-response curves for AMP, CIP,

NIT; extract eMIC at the chosen normalized-conductivity threshold; regress against E-test/broth-microdilution MIC and report R^2 .

Target: eMIC within $\pm$ one two-fold dilution of reference MIC and $R^2 > 0.9$. If not met, revisit frequency choice, medium conductivity, or add cell pre-concentration.

Escalation triggers. If time-to-result must drop below ~ 30 min, add membrane-filter or centrifugal cell pre-concentration (cheaper than picoliter microfluidics) before considering a microfabricated chamber. If baseline drift dominates, switch fully to 4-wire measurement and improve temperature control before spending on better electrodes.

Standardization. Always run in biological triplicate with paired growth and sterility controls; keep a CAMHB reference arm so calls remain traceable to CLSI/EUCAST; log temperature continuously and correct for the $\sim 2\%/^{\circ}\text{C}$ conductivity coefficient.

Caveats

- **Attribution correction:** The "electrical-MIC (eMIC)," 100-nL measurement volume, normalized-conductivity threshold, and strong 24-h broth-microdilution correlation come from **Spencer et al., ACS Sensors 2023 ("Electrical Broth Micro-Dilution")**, not from Scherer et al. 2021 (Lab on a Chip), which is a single-bacterium digital-impedance multiplexer proof-of-concept. [rsc](#) Cite accordingly.
- **Swami et al. 2022 specifics partially unverified:** The exact numeric susceptibility cutoff, measurement frequency, MIC R^2 , and LCGB conductivity value could not be verified from open sources (Elsevier paywall/robots block); the LCGB is described as a glucose/NaCl/HEPES zwitterionic mix and the method achieves 20 min (death-based) to 60–80 min (growth-based) AST at 2×10^5 CFU. Confirm specifics from the primary PDF before finalizing threshold numbers.
- **Electrode pricing is quote-only:** Metrohm DropSens does not publish list prices for G-IDEAU10/G-IDEAU5 (sold in packs of 20 –

the "5" and "10" denote 5 μm and 10 μm line/gap spacing, not quantity); the per-unit figures above are budgetary estimates, not confirmed quotes. Request a quote from Metrohm DropSens or nLab.

- **Impedance vs pH baseline dynamic-range difference:** Because impedance sensitivity scales as $1/\sigma$, media choices ideal for the paper's unbuffered-pH scheme (0.9% NaCl) are suboptimal for impedance; do not copy the paper's medium directly — minimize ionic strength instead.
- **Speed/resolution tradeoff:** The impedance prototype will generally not match the LOSC's 20-min, single-cell performance; realistic time-to-result is 30 min–3 h depending on medium, cell density, and antibiotic mechanism.
- **Bacteriostatic vs bactericidal:** Death/ion-release-based fast readouts apply mainly to bactericidal, surface-acting drugs; growth-based readouts are slower but general — plan the antibiotic panel accordingly.

- All component prices are as observed at the time of research (mid-2026) at DigiKey/Mouser and are subject to change and stock lead times (AD eval boards showed ~10-week standard lead times).
[digikey](#) [digikey](#)